

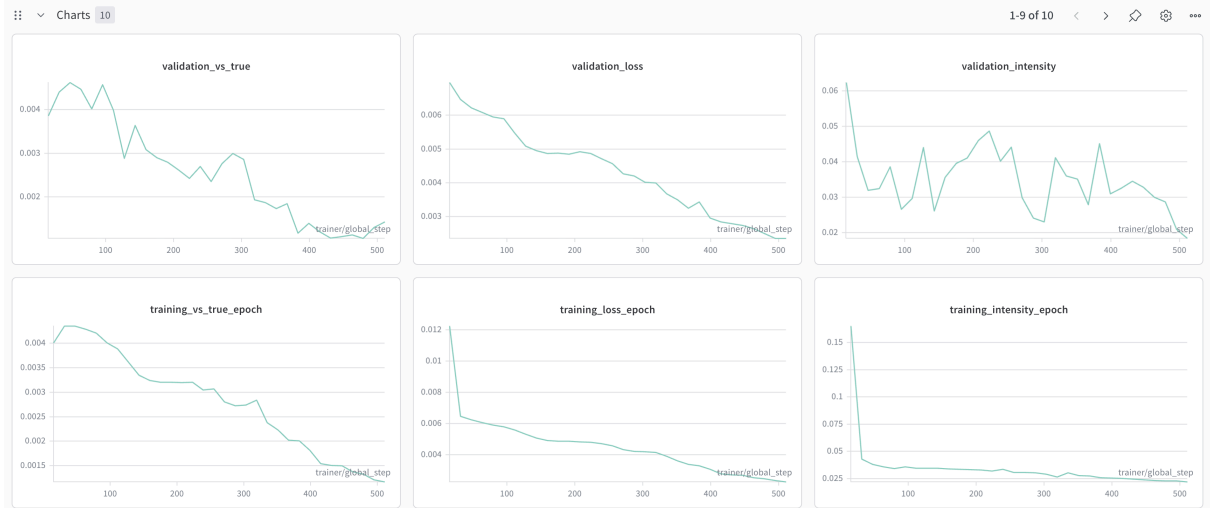


Figure 1: Loss charts after 32 epochs with a limited dataset of 256 images, resized at dimensions  $720 \times 405$ .

## Experiments

All experiments are conducted on a Macbook M1 with local M1 MPS chips. The results of training the final model for competition is compiled in [this Wandb report](#).

Overall, the training results illustrate the model's ability to capture the regions containing the players. The decreasing aggregate training and validation losses after 32 epochs indicate that the model can be further trained with more epochs and a biggest dataset to achieve better results.

Inspecting the model's outputs gives us a hint on what the model struggles to predict. Two main problems arise through inspection:

- The model struggles to 'find' out players further to the camera than those nearer. Not only does the size of the players on the camera decreases significantly, the players may blend into the background, which either are cropped out by `masking_unet`, or are unable to be detected by `heatmap_unet`.
- The model tends to produce false positives, mostly on the field lines and on the intersection of field lines.



Figure 2: False positive pixels on the boundary between the field area and the spectators' seating area.

We also examine our diagnostics by running a full inference pipeline.

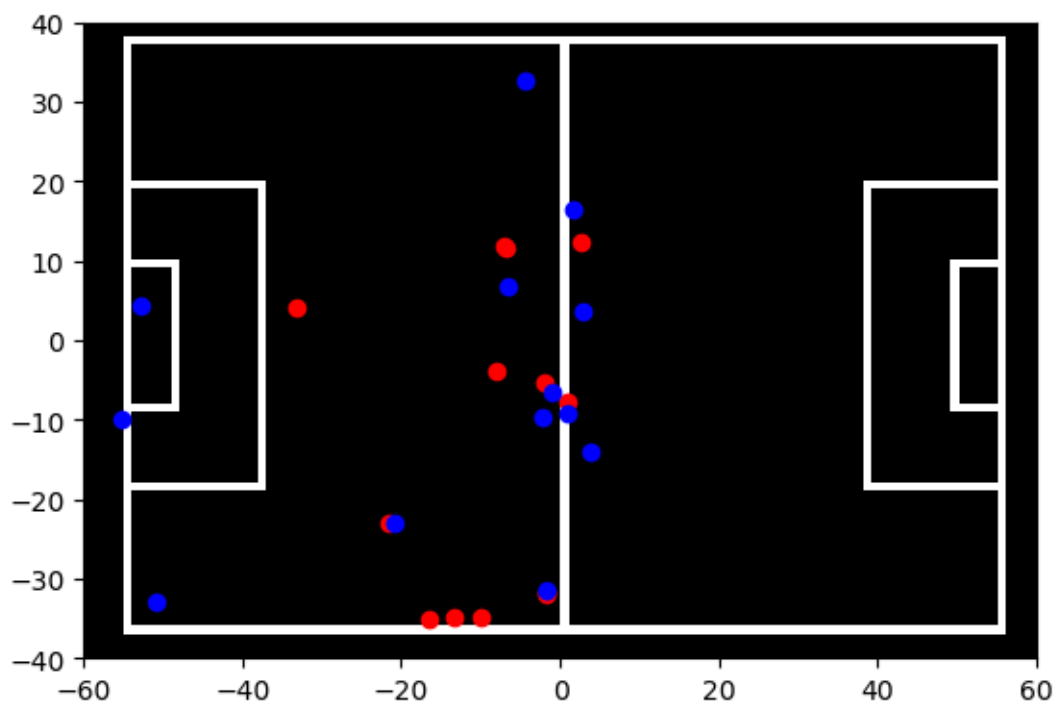
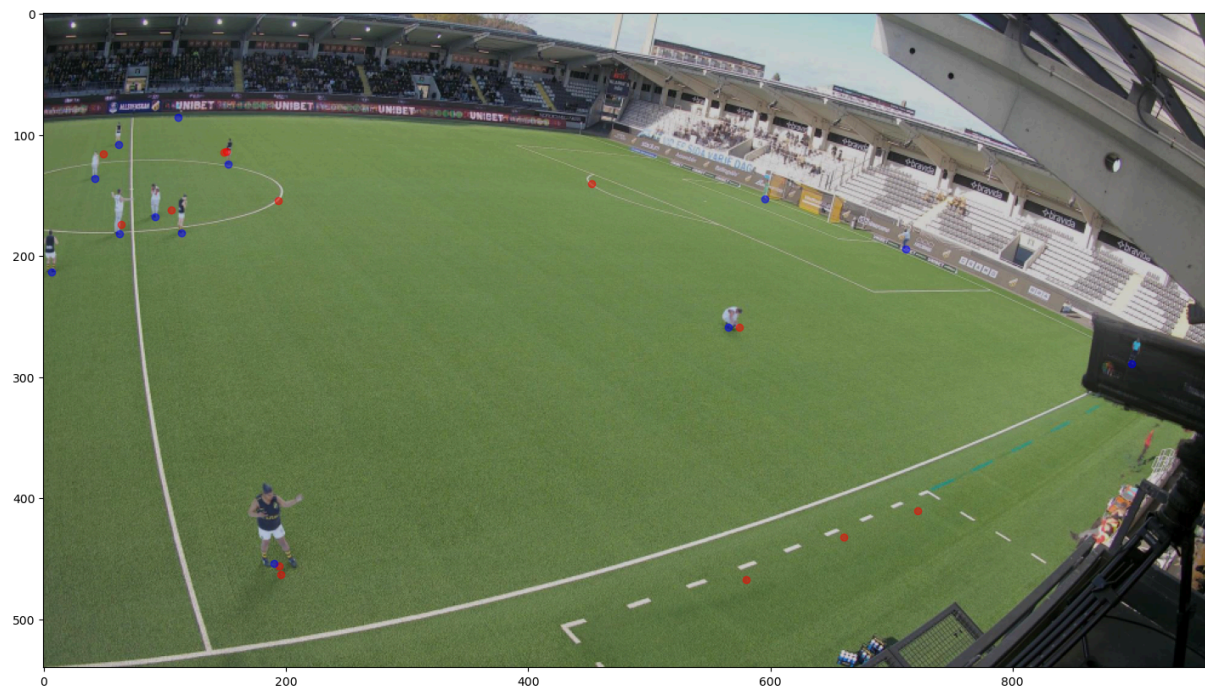


Figure 3: